AS343AI

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RV COLLEGE OF ENGINEERING®
Autonomous Institution affiliated to VTU
IV Semester B. E. Examinations
Aerospace Engineering
Model Question Paper
AS343AI-Aerospace Propulsion

Time: 03 Hours **Maximum Marks: 100**

Instructions to candidates:
1. Answer all the questions from PART A.
2. In PART B, Question 2 is compulsory. Answer any one full question from Question 3 or 4, 5 or 6, 7 or 8 and 9 or10. Each question carries 20 marks.

		PART-A	
1	1.1	In a centrifugal compressor, the process of energy transfer occurs in the ________.	01
	1.2	A solid propellant grain is obtained by mixing ammonium per chlorate and polyvinyl chloride. The role of Ammonium per chlorate in the propellant is ____________.	01
	1.3	The diameter of the propeller of an aircraft is 2.5m. It flies at a speed of 500kmph at an altitude of 8000m. For a flight to jet speed ratio of 0.7, determine Thrust produced. Take air density at an altitude of 8000m as $0.525 kg/m^3$.	02
	1.4	Propulsive power of a jet engine is defined as __________________ and given by the mathematical expression ______________.	01
	1.5	A liquid propelled rocket is generating a maximum thrust of 62250N when the mass flow rate of the nozzle is 24.88kg/sec. Determine the effective exhaust jet velocity of the nozzle when the chamber pressure is 7MPa and the throat and exit diameters of the nozzle are 8.55cm and 27.03cm respectively.	02
	1.6	ISRO needs to inject an INSAT Series satellite weighing 1250kg into the geosynchronous orbit through a rocket weighing 4700kgs. A delta V of 4000m/s is achieved by the nozzle to escape the earth's orbit. The engine mass flow rate is 9.5 kg/s with an effective velocity of 3,250 m/s. Calculate the burn time duration of the rocket.	02
	1.7	Consider a jet engine of an aircraft capable of generating 100kN of dry thrust at design conditions. The maximum pressure ratio and the turbine inlet temperature of the engine is 4:1 and 1000^0C respectively. If the maximum temperature of the combustion chamber is increased to 1100^0C, the specific power output of the engine for the same maximum compressor ratio will ___________.	02
	1.8	A rocket travelling at 6000m/s has a specific impulse of around 300 seconds. Calculate the effective jet velocity of the rocket.	01
	1.9	If the supersonic diffusion takes place between the leading edge of the forebody and the cowl lip, then such type of inlets are known as ______________.	01
	1.10	In general, a ___________ feed system gives a vehicle performance superior to a ___________ system when the total impulse or the mass of propellant is relatively low, the chamber pressure is low, the engine thrust-to-weight ratio is low.	01
	1.11	A normal shock inlet is geometrically much similar to a pitot type inlet, except for the lip being made very sharp. Apparently, if the lip of the normal shock inlet used for supersonic mission is made very thick, what will be the resulting phenomenon.	01
	1.12	By augmenting a Reheat system to simple gas turbine engine, ________________	01

		can be obtained.	
	1.13	Distinguish between restricted and unrestricted burning.	02
	1.14	For a turbojet engine operating on an ideal Brayton cycle, the pressure ratio is 10:1, the inlet temperature is 300 K, and the maximum cycle temperature is 1600 K. Calculate the thermal efficiency of the cycle.	02
		PART-B	
2	a	With the help of neat sketches, explain the working of a turboprop engine and also list their general characteristics.	08
	b	Illustrating the P-V and T-s diagrams, explain the causes for deviation of a Brayton cycle from the Ideal case.	08
3	a	Considering a control volume of a jet engine, derive an expression for the basic thrust in terms of effective speed ratio.	10
	b	A turboprop aircraft is flying at 600km/h at an altitude where the ambient conditions are 0.458 bar and -150C. Compressor pressure ratio 9:1 and maximum gas temperature 1200K. The intake duct efficiency is 0.9 and total head isentropic efficiency of compressor and turbine are 0.89 and 0.93 respectively. Calculate the specific power output in kJ/kg, thermal efficiency of the unit taking mechanical efficiency as 98% and neglecting the losses other than specified. Assume that exhaust gases leave the aircraft at 600km/h relative to aircraft.	06
		OR	
4	a	A turbojet power plant uses aviation kerosene having a calorific value of 43 MJ/kg. The fuel consumption is 0.18 kg/h/N of thrust, when the thrust is 9 kN. The aircraft velocity is 500 m/s the mass of air passing through the Compressor is 27 kg/s. Calculate the air-fuel ratio and overall efficiency.	08
	b	Define the following, also give their mathematical expressions : 1. Propulsive Efficiency 2. Thermal Efficiency 3. Specific Thrust 4. Overall Efficiency	08
5	a	Identifying the various zones in a combustion chamber, explain its working principle illustrating on a neat diagram.	08
	b	Based on the position of the normal shock wave, explain the different modes of operation of a Ramjet engine Inlet.	08
		OR	
6	a	What are nozzles? With appropriate sketches, explain the working of a convergent nozzle and also show the variation of pressure ratio with mass flow rate of the fluid passing through the nozzle.	10
	b	Compare the pressure and velocity variations in an Impulse and Reaction turbine and also bring out the differences between them.	06
7	a	Explain with neat sketches, the working of solid propellant rockets. Also mention their merits and demerits.	08
	b	Explain in detail, the working of turbo pump liquid propulsion feed system.	08
		OR	
8	a	Defining thrust profile, explain the different types of thrust profiles employed for rocket propulsion systems.	08
	b	Give the detailed classification of rocket propulsion systems.	08
9	a	a) Explain the following rocket performance parameters and give their mathematical expressions a) Specific Impulse b) Total Impulse c) Thrust Coefficient d) Impulse to weight ratio	08
	b	b) A rocket has the following data: 1. Propellant Flow Rate=5kg/s 2. Nozzle exit diameter=100mm 3. Nozzle exit pressure=1.02bar	08

		4. Ambient pressure=1.013 bar 5. Thrust chamber pressure=20bar 6. Thrust =7kN Determine the effective jet velocity, actual jet velocity, specific impulse and SFC.	
OR			
10	a	Write a short note on Specific Impulse and Specific Fuel Consumption of a rocket engine.	06
	b	a) A rocket projectile has the following characteristics: 1. Initial Mass=200 kg 2. Mass after rocket operation=130kg 3. Payload, non-propulsive structure etc=110kg 4. Rocket operation duration=3 sec 5. Average specific impulse of propellant=240sec Determine a) Mass ratio of the vehicle b) Propellant mass fraction c) Propellant flow rate d) Thrust e) Thrust to weight Ratio	10

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PART-A

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	L2	L3	L2	L2	L3	L1	L2	L2	L3	L3
COs	CO2	CO1	CO2	CO4	CO3	CO2	CO1	CO2	CO3	CO3
Q No	1.11	1.12	1.13	1.14	1.15	1.16	1.17	1.18	1.19	1.20
B T	L1	L3	L2	L2						
COs	CO1	CO3	CO2	CO2						

PART-B

Question No		B T Levels	Cos addressed		Question No		BT Levels	Cos addressed
2	a	L1	L3		3	a	L1	CO1
	b	CO1	CO3			b	L2	CO2
	c	L1	L3			c		
	d	CO1	CO3			d		
4	a	L1	L3		5	a	L2	CO2
	b	CO1	CO3			b	L3	CO3
	c	L1	L3			c		
	d	CO1	CO3			d		
6	a	L1	L3		7	a	L3	CO3
	b	CO1	CO3			b	L2	CO3
	c	L1	L3			c		
	d	CO1	CO3			d		
8	a	L1	L3		9	a	L2	CO3
	b	CO1	CO3			b	L3	CO4
	c	L1	L3			c		
	d	CO1	CO3			d		
10	a	L1	L3		11	a		
	b	CO1	CO3			b		
	c					c		
	d					d		

Signature of Scrutinizer: Signature of Chairman

Name: Name: